THE LAGRANGIAN METHOD FOR OPTIMIZATION

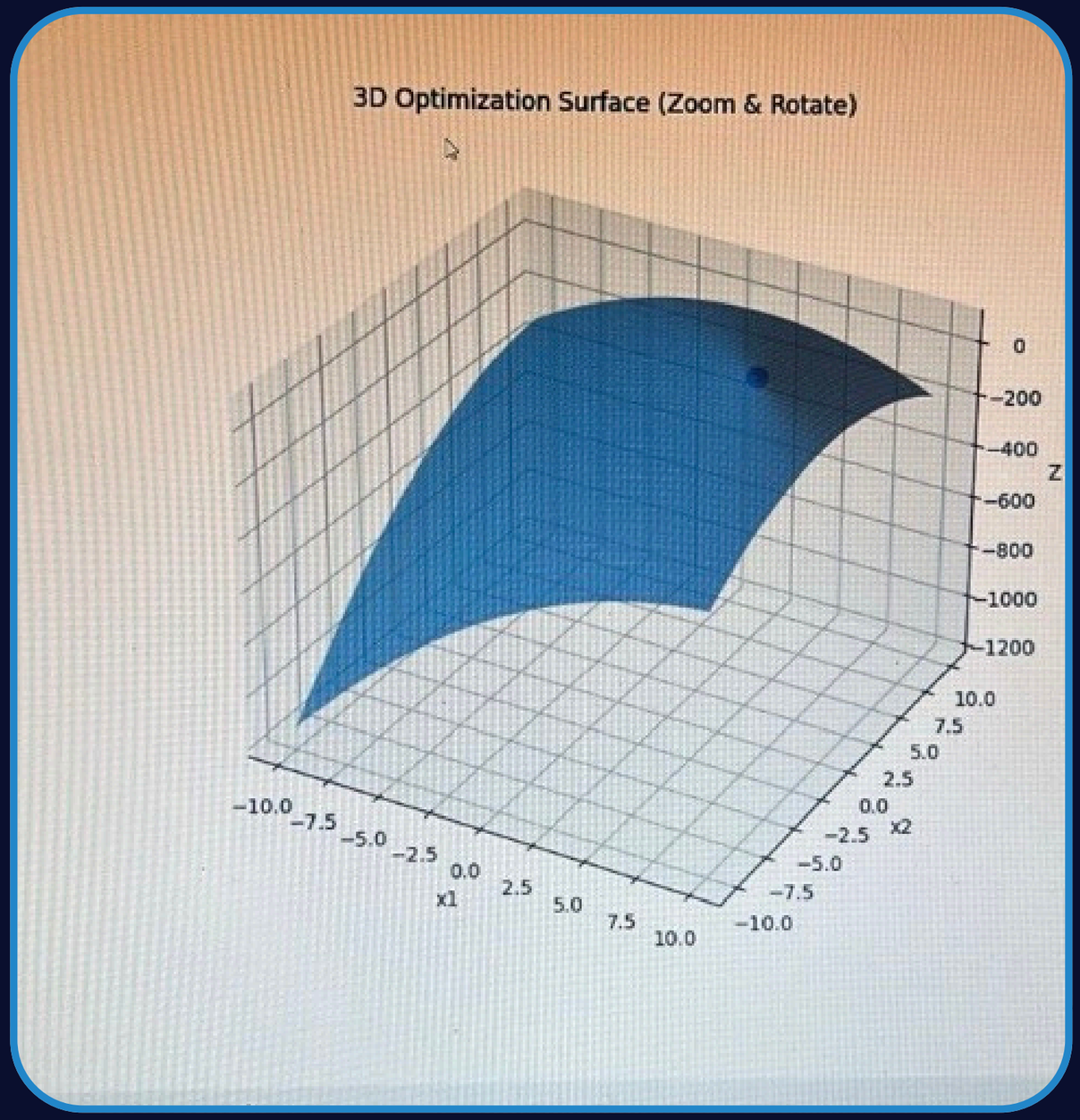
The Lagrangian Method is used to solve nonlinear optimization problems with equality constraints.
It converts a constrained problem into an unconstrained problem using a multiplier (λ).

THE LAGRANGIAN METHOD IS USED TO SOLVE NONLINEAR OPTIMIZATION PROBLEMS WITH EQUALITY CONSTRAINTS.
IT CONVERTS A CONSTRAINED PROBLEM INTO AN UNCONSTRAINED PROBLEM USING A MULTIPLIER (Λ).

FORMULA
LAGRANGIAN FUNCTION:

$$L = F - \Lambda G$$

$$\nabla F = \Lambda \nabla G$$



FEATURES
- GUI-BASED INPUT
- AUTOMATIC SOLUTION
- STEP-BY-STEP OUTPUT
- 3D GRAPHICAL VISUALIZATION

TECHNOLOGY USED
- PYTHON
- SYMPY (SYMBOLIC SOLVING)
- TKINTER (GUI)
- MATPLOTLIB (GRAPH)

CONCLUSION:

- LAGRANGIAN METHOD SIMPLIFIES CONSTRAINED OPTIMIZATION PROBLEMS
- PROVIDES ACCURATE AND EFFICIENT SOLUTIONS
- WIDELY USED IN ENGINEERING AND MATHEMATICS

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